Entity Relationship Diagram

I. ER Diagram Textual Representation

This ER diagram represents the core entities for the Flood Detection System, outlining data flow from environmental sensors to prediction alerts.

Main Entities and Attributes

1. Sensor

Description: Represents physical sensors deployed for monitoring.

Attributes:

SensorID (PK)

Type (e.g., Water Level, Rain Gauge)

InstallationDate

Status (e.g., Active, Inactive, Maintenance)

StationID (FK)

2. Monitoring Station

Description: A physical location containing one or more sensors.

Attributes:

StationID (PK)

StationName

Location (Latitude, Longitude)

RiverBasinName

Elevation

3. Sensor Reading

Description: Time-stamped data generated by a specific sensor.

Attributes:

ReadingID (PK)

SensorID (FK)

Timestamp

Value (e.g., in meters, millimeters)

4. Prediction Model

Description: The algorithm used to forecast potential flood events.

Attributes:

ModelID (PK)

ModelName

AlgorithmType

LastTrainedDate

AccuracyMetric

5. Flood Prediction

Description: The outcome of running the prediction model on input data.

Attributes:

PredictionID (PK)

ModelID (FK)

StationID (FK)

PredictionTime

ForecastTimeWindow

PredictedFloodLevel

ProbabilityScore

6. Alert

Description: An alert generated when a flood prediction exceeds safety thresholds.

Attributes:

AlertID (PK)

PredictionID (FK)

TriggerTime

SeverityLevel (e.g., Low, Medium, High, Critical)

Status (e.g., Active, Resolved)

7. System User (Admin/Monitor)

Description: Personnel managing the system or receiving direct alerts.

Attributes:

UserID (PK)

UserName

Role (e.g., Admin, Hydra-Meteorologist, Emergency Dispatcher)

Email

ContactNumber

II. Entity Relationships

Based on the system requirements and logical flow, the relationships are defined below:

Monitoring Station \rightarrow Sensor: One Station can host multiple Sensors. (One-to-Many, 1:M)

Sensor \rightarrow Sensor Reading: One Sensor generates multiple Readings over time. (One-to-Many, 1:M)

Prediction Model \rightarrow Flood Prediction: One Model can generate multiple Predictions. (One-to-Many, 1:M)

Monitoring Station \rightarrow Flood Prediction: A Prediction is specific to a Monitoring Station (based on its inputs). (One-to-Many, 1:M)

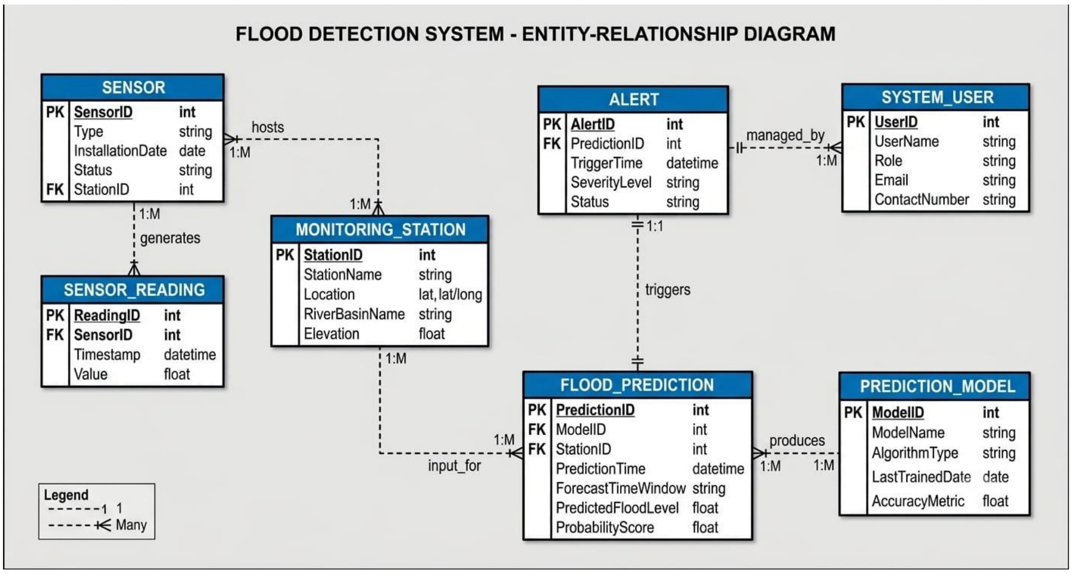
Note: This relationship is a critical data association for the model.

Flood Prediction \rightarrow Alert: A critical Prediction generates exactly one Alert (when thresholds are met). (One-to-One, 1:1)

Alert \rightarrow System User: One Alert can be managed by one User, and a User can manage multiple alerts (during their shift). (One-to-Many, 1:M)

Alternatively, if many users are notified: (Many-to-Many, M:N) via a junction table AlertNotification (AlertID, UserID, NotificationTime). I have used the simpler 1:M relationship for direct management, matching the reference structure.

III. Visual ER Diagram



IV. Entity Explanations and Design Notes

1**. Monitoring Station & Sensor:** A crucial part of physical data collection. It is highly structured, mapping multiple sensor types (water level, rain) back to a single station (a physical asset). This normalization prevents data redundancy.

2. Sensor Reading: The system's primary time-series data source. This table will be massive in a real application. It connects directly to the SensorID (FK).

3. Prediction Model: Separating the model definition allows the system to compare different algorithms or manage updates without altering the prediction table.

4. Flood Prediction: This entity is the system's analytical engine output. It connects the inputs (StationID) with the processing logic (ModelID). This structure mirrors the prediction architecture in your sample reference, just applied to floods rather than loans.

5. Alert: Alerts are derived events triggered by specific critical predictions. A critical prediction (FloodPrediction) results in one unique Alert. This 1:1 relationship ensures accountability and a clear audit trail.

6. System User: Focuses on roles directly involved in the system's operation (Admins, Forecasters) and decision-making (Dispatchers).